

Foundation Mathematics 1017SCG
Week 6 Workshop Questions

1. The number of fish in a small pond can be modelled using $N = N_0e^{kt}$ where N is the number of fish in the pond t months after the start of the year and N_0 is the number of fish in the pond at the start of the year. There were 80 fish in the pond at the start of the year.
 - (a) If there was 100 fish in the pond at the start of April, find the value of the growth factor k to four decimal places.
Hint: It is important to note that in this question $t = 0$ is the start of January, $t = 1$ is the start of February, $t = 2$ is the start of March.
 - (b) Using the model, predict the number of the fish in the pond at the start of August.
 - (c) When would there be 220 fish in the pond?
2. The amount of radioactive substance remaining in a small container can be modelled using $N = N_0e^{kt}$ where N is the mass of radioactive substance remaining (measured in grams) and t is the number of years that has passed. There was initially 15 grams of radioactive substance in the container, but only 10 grams 60 years later.
 - (a) Calculate the decay factor k to four decimal places.
 - (b) Determine the half-life of the radioactive substance. The half-life is the time it takes for half of the radioactive substance to decay.
3. You have been asked to model Australia's population using an exponential growth model of the form $N = N_0e^{kt}$ where N is Australia's population (in millions) and t is the number of years since the start of the year 2000. Australia's population at the start of 2000 was approximately 19 million.
 - (a) If Australia's population at the start of 2021 was 25.7 million, find the growth factor k for the exponential growth model. Round your answer to **six** decimal places.
 - (b) Using the model, predict Australia's population at the start of 2030. Round your population prediction to the nearest ten thousand.
 - (c) According to the model, in what year is Australia's population expected to reach 40 million?
4. *In this question, the exponential growth model has been written with 10 as the base (rather than using e as the base). You may find logarithms with base 10 are easier to use with this model than logarithms with base e .*
The population, P , of a certain variety of fish introduced into a dam can be modelled using $P = 400 \times 10^{0.08t}$ where t is the number of months after the introduction of fish to the dam.
 - (a) Using the model, determine the number of fish in the pond after 3 months.
 - (b) Using the model, determine the number of fish in the pond after 8 months.
 - (c) How long will it take for the number of fish in the dam to reach 5000?
 - (d) What is the minimum number of fish in the dam? When does this occur?

Question 5-9 involve solving trigonometric equations which require using the special triangles and unit circle. As it is difficult to include these diagrams in the solutions, it is best to ask questions and receive help for these questions during the weekly workshop.

5. Solve the following equations where $0^\circ \leq \theta \leq 360^\circ$.

(a) $\sin(\theta) = \frac{1}{2}$

(b) $\tan(\theta) = \frac{1}{\sqrt{3}}$

(c) $\cos(\theta) = \frac{1}{\sqrt{2}}$

(d) $\sin(\theta) = \frac{\sqrt{3}}{2}$

6. Solve the following equations where $0 \leq \theta \leq 2\pi$. Give your answers as exact values.

Hint: You may find it helpful to rearrange the trigonometric equation.

(a) $\tan(\theta) = \sqrt{3}$

(b) $10 \cos(\theta) = 5$

(c) $\sqrt{2} \sin(\theta) = 1$

(d) $\sin(\theta) = \cos(\theta)$ Hint: $\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$

7. Solve the following equations where $0^\circ \leq \theta \leq 360^\circ$.

(a) $\sin(\theta) = -\frac{1}{2}$

(b) $\cos(\theta) = -\frac{\sqrt{3}}{2}$

8. Solve the following equations where $-\pi \leq \theta \leq \pi$.

(a) $\tan(\theta) = -\frac{1}{\sqrt{3}}$

(b) $\cos(\theta) = -\frac{1}{\sqrt{2}}$

9. Solve the following equations where $\pi \leq x \leq 2\pi$. Round your answers to two decimal places.

*Hint: You will not be able to use the special triangles in this question. Using the trigonometric inverse functions on your calculator, find **one** solution to the equation and then determine which quadrants may also contain solutions.*

(a) $5 \tan(x) - 2 = 0$

(b) $10 \cos(x) + 9 = 0$

10. Consider the function $f(x) = 2x - 5$
- (a) Evaluate $f(6)$
 - (b) Evaluate $f(-3)$
11. Consider the function $g(x) = -x^2 + 3x$
- (a) Evaluate $g(4)$
 - (b) Evaluate $g(-2)$
12. Consider the function $f(x) = 3x^2 + 2x - 1$.
- (a) Evaluate $f(4)$.
 - (b) Evaluate $f(-5)$.
 - (c) Evaluate $f(0)$.
 - (d) Find $f(3a)$. Expand and simplify your answer.
 - (e) Find $f(b + 1)$. Expand and simplify your answer.
13. Consider the function $f(x) = \sqrt{x}$.
- (a) Explain why $x = -3$ is **not** included in the domain of the function.
 - (b) Explain why $x = 4$ is included in the domain of the function.
 - (c) What is the domain of the function $f(x)$?
14. Consider the function $f(x) = \frac{1}{x - 2}$.
- (a) Explain why $x = 2$ is **not** included in the domain of the function.
 - (b) Explain why $x = 5$ is included in the domain of the function.
 - (c) What is the domain of the function $f(x)$?
15. Consider the function $f(x) = \log_{10}(x)$.
- (a) Explain why $x = 0$ is **not** included in the domain of the function.
 - (b) Explain why $x = -10$ is **not** included in the domain of the function.
 - (c) Explain why $x = 1$ is included in the domain of the function.
 - (d) What is the domain of the function $f(x)$?
16. Consider the function $f(x) = x^2$.
- (a) Explain why $f(x) = -9$ is **not** included in the range of the function.
 - (b) Explain why $f(x) = 16$ is included in the range of the function.
 - (c) What is the domain and range of the function $f(x)$?